Node Listen Manual
Receive and transmit audio, setting by setting

Node Listen puts one node in your phone's browser: listen with about half a second of delay, talk with push-to-talk, and connect favorites. This manual follows the audio both ways, from the radio to your ear and from your mic to the air, and explains every setting along the way: where it acts, what it does, when to change it, and how to measure the result with the built-in meters, tests and Audio check.

RECEIVE TRANSMIT SETTINGS LEVELS TROUBLESHOOTING

COVERED BY THIS EDITION

TOOL	FILE	VERSION
Node Listen	`nodelisten_v0_4_17_20261020.py`	0.4.17
Dashboard	`nodecontrol_pi02w_v9_3_93` / `nodecontrol_v9_3_89`	9.3.93-pi02w · 9.3.89 (full), for M17 favorites and the page following Node Listen

Node Listen is a common tool: the same file runs on the Pi Zero 2 W and the full build. The rest of the suite is covered in the user manual, `docs/ASL-DVS_User_Manual.pdf`.

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How to read it

If you only want to listen, chapters 2 and 3 are enough. If you transmit, read chapters 5 and 7 before your first contact: most audio complaints come from levels that were never set. Chapter 6 is the reference to come back to. Each setting is marked with where it is kept:

MARK	MEANING
NODE	Kept on the Pi in `/etc/asl_dvs/nodelisten.conf`. Changes it for everyone who uses Node Listen on this node. Needs you to be logged in.
BROWSER	Kept in this browser only. Each phone, tablet or computer has its own, and changing it on one leaves the others alone. No login needed.

> **NOTE** Levels are given in **dBFS**: decibels below digital full scale. 0 dBFS is the loudest a sample can be; −6 dBFS is half that; −20 dBFS is a tenth. Speech is measured as an average (what you hear as loudness) and as a peak (the loudest instant, which is what clips).

PART I

Getting going

What Node Listen does, how to set it up on a node, and a tour of the page.

1. What Node Listen is
2. Install and first setup
3. The page

What Node Listen is

A web page served by the Pi on port 8993 that lets a phone, tablet or computer listen to one node, transmit through it, and connect a short list of favorites. There is no app to install.

RECEIVE
The node's audio, about half a second behind, cleaned up on the Pi (CTCSS and hum cut, levels evened out, squelch tails muted) and played in the browser with a live spectrum.

TRANSMIT
Push-to-talk from the phone's microphone, processed in the page (filter, leveler, noise gate) and sent to a private node, 1921, that Node Listen keeps linked to your node.

CONNECT
Up to 10 favorites: AllStar nodes, EchoLink stations and M17 reflectors. Node Listen reports whether each link actually came up.

MEASURE
Live meters, a mic test, a test through the node, and an Audio check that watches a few minutes of real use and says what to change.

1.1 What it needs

ITEM	WHY
An ASL3 node (Asterisk with app_rpt)	Node Listen takes the node's audio from Asterisk and transmits through it.
Python 3.11 or later	As on Debian 12 and 13. Node Listen uses only the standard library.
A modern browser	Chrome, Firefox, Safari or Edge from the last few years. The microphone needs the `https://` page.
Optional: the Dashboard (Node Controller)	Needed for M17 favorites; when it runs, connects from its node go through it so it shows them.

1.2 How it fits on the node

Node Listen adds two things to `/etc/asterisk/rpt.conf`, each with a backup first: one `outstreamcmd` line in your node's stanza, which sends the node's audio to Node Listen, and a private node **1921**, which carries your transmissions in. Everything else stays as it was: the radio, the other links and DVSwitch hear and send exactly what they did before.

Install and first setup

Ten minutes from nothing to a first test transmission.

2.1 Install

1. The installer (7.2.0 and later) asks whether to add Node Listen; instmon (2.58.0 and later) lists it with the other tools and updates it. By hand: `sudo python3 nodelisten_v….py --install`.
2. Open `http://<pi-address>:8993`. Node Listen moves you to `https://`. The certificate is made on the Pi, so the browser warns once: accept it.
3. Log in with the Pi's root password.

2.2 Receive: send the node's audio here

1. Setup tab, open the **Receive** card. The table lists each node in `rpt.conf` and its `outstreamcmd`, if any.
2. Pick your node and press **Send its audio here**. That node becomes Node Listen's main node.
3. Press **Restart Asterisk** (it asks first; every link drops and reconnects).
4. Listen tab: within a few seconds the status line shows a green **feed** light, and audio plays when the node passes any.

> **CAREFUL** A node takes one `outstreamcmd`. If yours already has one (a Broadcastify feed, for example), Node Listen refuses rather than replace it.

2.3 Transmit: add node 1921

1. Setup tab, open the **Transmit** card and press **Add node 1921**. Node Listen adds a `[1921]` stanza (your node's template, a USRP channel, no hang time, telemetry or courtesy tone) and a `[nodes]` line, after a backup.
2. Press **Restart Asterisk**. Within 20 seconds the card's summary reads *Set up and linked*: Node Listen links 1921 to your node and relinks it whenever it drops.

2.4 First test transmission

1. Setup tab, **Microphone** card: press **Test mic**, talk for 4 seconds, and read the report (section 8.2). Adjust **Mic gain** until it says *Good*.
2. Press **Test through the node**. It transmits for 4 seconds, over the air and to every link, and plays back what the node sent out.
3. Listen tab: hold the push-to-talk button and make a short call.

The page

Three tabs along the bottom: Connect, Listen and Setup. The title bar and the tabs bar stay put; only the middle scrolls, and the Listen tab is sized to fit the screen without scrolling.

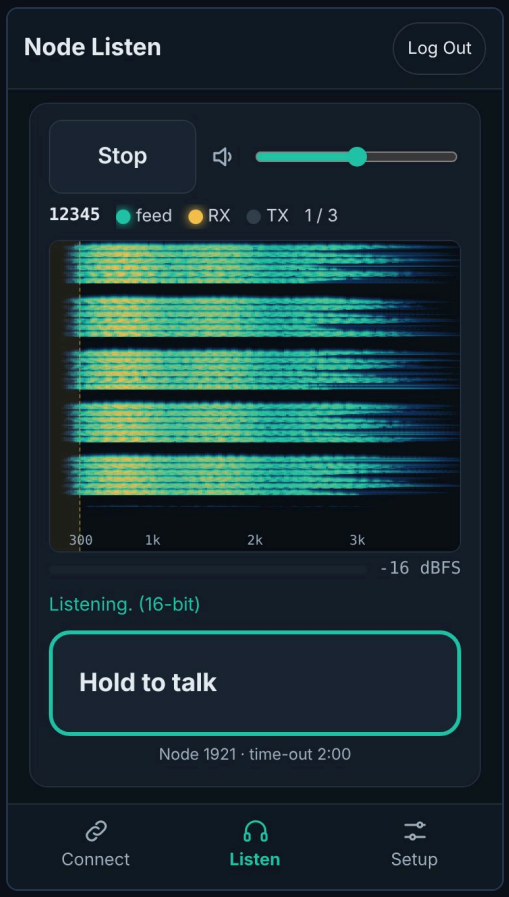


Figure 3.1 · Screenshot (test signal). The Listen tab.

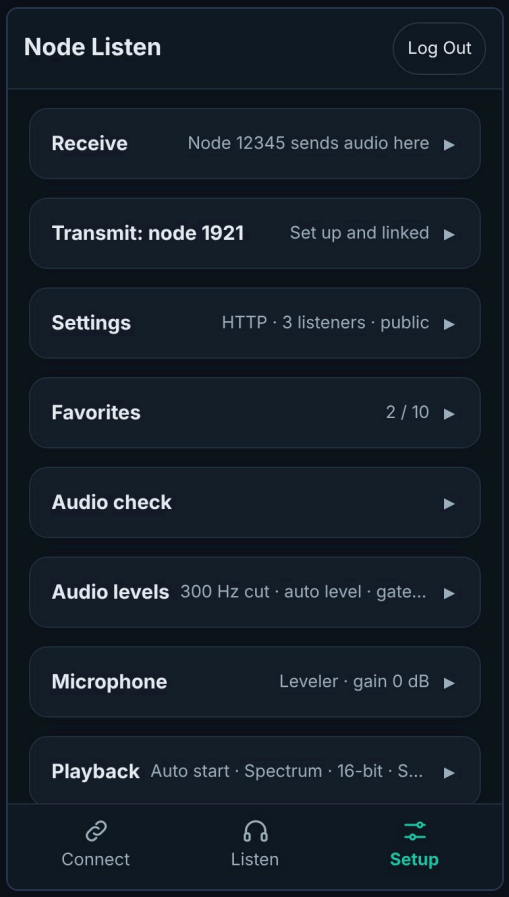


Figure 3.2 · Screenshot. The Setup tab: each card closed, with its summary.

3.1 Title bar and login

PART	WHAT IT DOES
Version	Hidden on narrow phones.
Node Controller	Opens the Dashboard, or switches to its tab if it is already open. Shown only while the Dashboard runs.
Log In / Log Out	The root password. Five wrong tries from one address lock that address out for 5 seconds, doubling up to 5 minutes. A login lasts 12 hours from the last use.

With **Listen without logging in** on (Settings card), anyone who can reach the page gets the Listen tab without the password but can't transmit or change anything.

3.2 Listen tab, top to bottom

PART	WHAT IT DOES
Listen / Stop and volume	Starts and stops the audio; the slider beside it is the volume, 0 to 200 %. Listening starts by itself when the page opens (section 6.7).
Status line	Node number · **feed** light (green while audio arrives from Asterisk; "no feed 12 s" otherwise) · **RX** light (the node is passing audio) · **TX** light (someone transmits from Node Listen) · listeners now / limit · `CHECK 2:13` while an Audio check runs. Hover over a light for details.
Spectrum	0 to 4 kHz, waterfall or bars; tap to switch (chapter 8).
Level bar	The node's level, or your mic level (red **TX**) while you hold push-to-talk.
Message line	*Listening.*, *Tap anywhere to start listening*, or why the stream stopped.
Push-to-talk	Hold to talk (chapter 5). Red **TALKING** with the time; amber **Busy: who 0:12** when someone else transmits.

3.3 Setup tab

Cards from one-time setup at the top to everyday use at the bottom: **Receive**, **Transmit**, **Settings**, **Favorites**, **Audio check**, **Audio levels**, **Microphone**, **Playback**, **Log**. Tap a title to open or close it; every card starts closed, showing a one-line summary such as *Node 41234 sends audio here* or *300 Hz cut · auto level · gate light*. Chapter 6 goes through them all.

3.4 Connect tab

The node's links at the top, with the linked favorites and **Disconnect**; below, the favorite to connect (chapter 9).

PART II

The audio, end to end

Where the sound goes, what changes it at each step, which setting acts where, and how to set and check levels.

4. Receive (RX) audio
5. Transmit (TX) audio
6. Settings reference
7. Getting levels right
8. Reading the spectrum and the meters

Receive (RX) audio

From the radio to your ear: everything the node hears, cleaned up once on the Pi for every listener, then buffered, played and drawn by each phone.

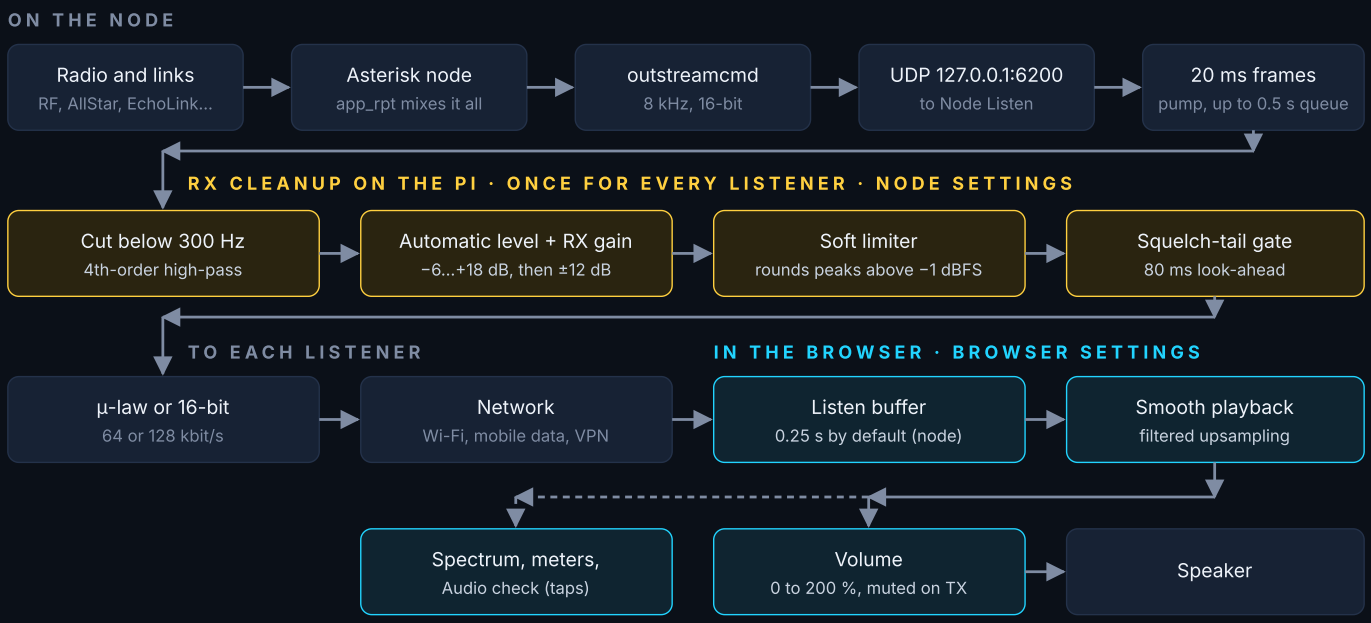


Figure 4.1 · The receive path. Amber steps run on the Pi once for everyone, blue steps on each listener's own device. The dashed branch is measuring only: the spectrum, the level bar and the Audio check see the audio before your volume.

4.1 From the node to Node Listen

app_rpt mixes everything your node hears (the radio, every link, telemetry) and writes it, 8,000 samples a second at 16 bits, to the `outstreamcmd` program, a small part of Node Listen that passes it on to `127.0.0.1:6200`. The audio pump cuts it into 20 ms frames (160 samples) and sends one to every listener every 20 ms. If audio arrives in bursts, up to 0.5 s waits in a queue; beyond that the oldest is dropped and counted.

The status line's **feed** light is green while frames arrive. The **RX** light and the busy lockout use the node's own level before any cleanup: a frame louder than −50 dBFS counts as audio, held for 0.6 s.

4.2 RX cleanup on the Pi

Three steps, done once for every listener, change only what Node Listen's listeners hear. They are on by default; chapter 6 has each setting in detail. Nothing is done while nobody listens, and near-silent frames (under −65 dBFS) skip the work.

STEP	WHAT IT DOES TO THE SOUND
Cut below 300 Hz	Removes hum and the lower CTCSS (PL) tones: 100 Hz drops by about 38 dB, 150 Hz by 24 dB, 250 Hz by 7 dB. Voice above 300 Hz is untouched.
Automatic level	Aims speech at about −20 dBFS: brings quiet stations up by as much as 18 dB and loud ones down by as much as 6 dB. It turns down within a frame or two, comes up at 3 dB a second (faster for the first half second of a new transmission), and holds still between transmissions. RX gain (Audio levels) is added on top.

Soft limiter	Up to about −1 dBFS nothing changes; above that, peaks are rounded off smoothly instead of clipped.
Squelch-tail gate	Mutes the burst of hiss after a station unkeys, and long silences. It tells hiss from voice by how alike neighbouring samples are: voice scores high, hiss low. To mute a tail from its first frame, it holds the audio back 80 ms.

4.3 To your phone and out of the speaker

- **Format:** 16-bit (128 kbit/s) by default, or µ-law (64 kbit/s) with **16-bit audio** off. µ-law is the phone-line format: half the data, a little more hiss on quiet audio.
- **Listen buffer:** the page plays each 100 ms of audio a fixed time after it arrives, 0.25 s by default, so a late packet doesn't make a gap. When audio arrives later than the buffer can cover, playback restarts (a *dropout*), and the Audio check counts it.
- **Smooth playback:** the page upsamples 8 kHz to the phone's own rate (44.1 or 48 kHz) with a proper filter, which removes the faint sizzle some phones add.
- **Volume** and then the speaker. The spectrum and meters see the audio before the volume.

4.4 Where the half second goes

STAGE	DELAY
Asterisk to Node Listen	a few tens of ms
Squelch-tail gate look-ahead	80 ms (only with the gate on)
Page chunks	up to 100 ms
Listen buffer	250 ms by default (150 ms to 1 s)
Network	a few ms on Wi-Fi at home; more on mobile data

The total is about half a second at home. This is why the node's own audio comes back late when you test through it, and why Node Listen ignores the busy lockout for 1 s after your own transmission.

Transmit (TX) audio

From your mic to the air: shaped in the page, sent in 20 ms pieces, smoothed on the Pi and fed to node 1921, which is linked to your node.

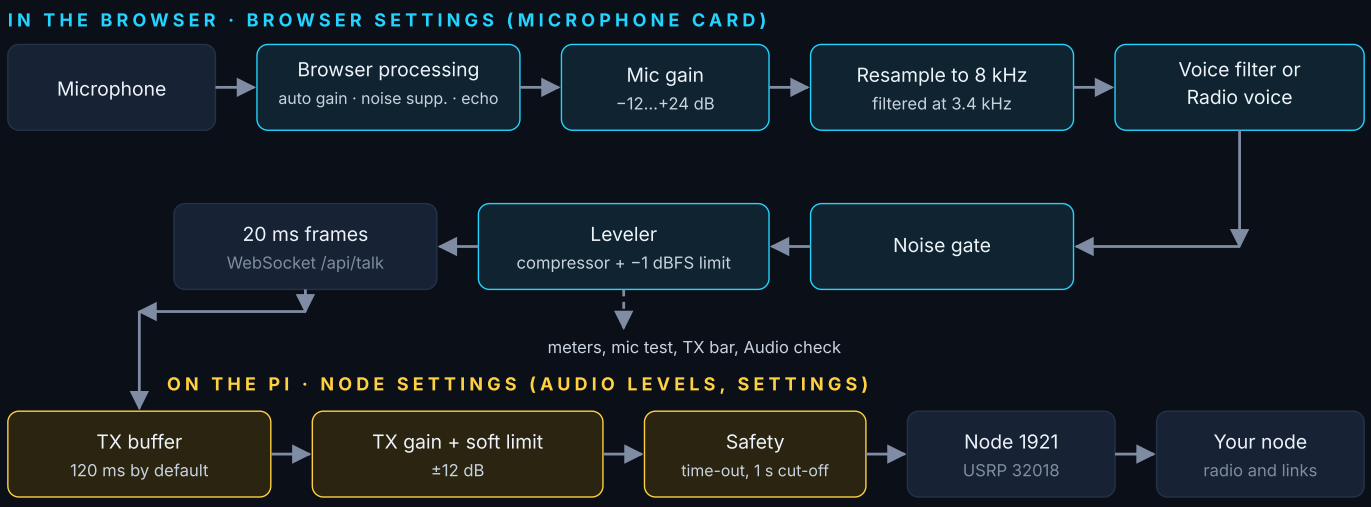


Figure 5.1 · The transmit path. Blue steps run in your browser with your device's settings, amber steps on the Pi with the node's settings. The meters measure right after the leveler, so they show what the Pi will get.

5.1 In the browser

1. **Browser processing.** The phone's own **Automatic gain**, **Noise suppression** and **Echo cancellation**, each switchable. They run before Node Listen sees the sound.
2. **Mic gain**, −12 to +24 dB: a plain volume control on the mic.
3. **Resample to 8 kHz** with a filter that cuts at 3.4 kHz, so sounds above it don't fold back as hiss or whine.
4. **Voice filter** (cuts rumble below about 250 Hz) or **Radio voice** (cuts below 300 Hz and lifts 2.5 kHz by 4 dB, a crisper, more "radio" sound). With Radio voice on, the voice filter is skipped.
5. **Noise gate** (Off, Light, Strong): turns the mic down between words.
6. **Leveler**: a gentle compressor that keeps your level even, followed by a limiter that never lets a peak past −1 dBFS.
7. Every 20 ms (160 samples) a frame goes to the Pi over the WebSocket `/api/talk`.

5.2 On the Pi and on the air

1. **TX buffer.** The Pi waits until it has this much of your audio (120 ms by default) before it starts sending, so short hiccups on the phone's connection don't make gaps. If it runs dry anyway, it sends silence, refills, and counts a *gap*.
2. **TX gain** (−12 to +12 dB, with a soft limit at about −1 dBFS), only with **Adjust audio levels here** on.
3. The frames go as USRP packets to node 1921 (`127.0.0.1:32018`), keyed. On release the last frame is faded out and three unkey packets follow.

4. Node 1921 is linked to your node, so your audio goes out on the radio and to every link, like a local user's.

5.3 The safety rules

RULE	WHAT HAPPENS
Busy lockout	With **Don't transmit while the node is busy** on, a press is refused while the node passes audio (the RX light), except within 1 s of your own unkey.
One at a time	A second person's press is refused while someone transmits from Node Listen.
Time-out	The transmission stops at **TX time-out** (120 s by default). Let go and press again.
Audio stopped	If no audio reaches the Pi for 1 s (a dropped connection), it unkeys.
Page hidden	Switching apps, locking the phone or leaving the page releases push-to-talk.

Settings reference

Every setting, card by card, in the order of the Setup tab. NODE settings need you to be logged in and apply to everyone; BROWSER settings are this device's own.

6.1 Receive card

CONTROL	WHAT IT DOES
Node table	Each node in `rpt.conf` and its `outstreamcmd`: *Sends audio here*, *to another UDP port (press Send again)*, another program's line, one inherited from a template, or *Not sending*.
Send its audio here	Adds Node Listen's line to the picked node (taking it off any other node), backs up `rpt.conf`, and makes that node the main node. Takes effect after Restart Asterisk.
Remove	Takes the line out again (with a backup).
Restart Asterisk · Refresh	Restart asks first; every link drops and reconnects. Refresh rereads `rpt.conf`.
Line added · Feeder · UDP input	The exact line; whether Asterisk is running the feeder program now; whether Node Listen is listening on the UDP port.

6.2 Transmit card

CONTROL	WHAT IT DOES
State	*Not set up*, *Set up and linked*, *Set up; linking*, *Asterisk does not know it yet: restart Asterisk*, or what is wrong.
Add node 1921	Adds the `[1921]` stanza and its `[nodes]` line (with your IAX port if it isn't 4569); also repairs a wrong `[nodes]` line. Refuses if `rpt.conf` has a `[1921]` it didn't add.
Remove	Takes both out (asks first). Push-to-talk stops working.

6.3 Settings card NODE

SETTING	DEFAULT	RANGE	WHAT IT DOES, AND WHEN TO CHANGE IT
Web port	8993	1024–65535	The page's port. Change only if something else uses 8993. Restarts Node Listen.
UDP port	6200	1024–65535	Where the node's audio arrives. Changing it rewrites the `outstreamcmd` line too; restart Asterisk.
Max listeners	3	1–10	How many people can listen at once. Each takes 64 or 128 kbit/s of upload on your internet line when they're away from home.
TX time-out (s)	120	30–600	The longest one transmission can last.
HTTPS	on	on / off	Needed for the microphone. Off only for a listen-only node. Restarts Node Listen.
Don't transmit while the node is busy	on	on / off	The busy lockout (section 5.3). Turn it off only if a constant noise on the node keeps it busy.

Listen without logging in	off	on / off	Anyone who can reach the page may listen (not transmit).

6.4 Audio check card BROWSER

Length (2, 5, 10 or 15 minutes; 5 by default), **Start check**, **Stop**, then the report with **Apply**/**Undo**, **Compare with last check** and **Copy report**. Section 7.4 explains it.

6.5 Audio levels card NODE

Adjust audio levels here

Off by default. Off, the four settings below are not used: no TX or RX gain, a 0.25 s listen buffer and a 120 ms TX buffer. On, they take effect. The RX cleanup further down works either way.

SETTING	DEFAULT	RANGE	WHAT IT DOES, AND WHEN TO CHANGE IT
Mic to node (TX)	0 dB	−12 to +12 dB	Gain on every transmission from Node Listen, on the Pi, for everyone. Use it when all phones come out too quiet or too loud on the air (the node test shows it). For one phone, use its Mic gain instead.
Node to phone (RX)	0 dB	−12 to +12 dB	Gain on what every listener hears, after Automatic level. Use it when audio is quiet even with Automatic level on, or when you turn Automatic level off.
Listen buffer	0.25 s	0.15, 0.25, 0.5, 1 s	How far behind real time the page plays. Larger rides over a weak connection without dropouts; smaller is closer to live. 0.15 s on good Wi-Fi, 0.5–1 s on mobile data.
TX buffer	120 ms	60, 120, 200, 300 ms	How much of your audio the Pi collects before sending. Raise it if transmissions have gaps (TX gaps, the Audio check); lower it for a slightly quicker start.

Below them: **TX peak** (the loudest point of the current or last transmission, after TX gain), **RX level** (the node's level, before cleanup) and **TX gaps**. Aim for TX peaks of −6 to −3 dBFS on normal speech.

RX cleanup

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Cut below 300 Hz	on	Removes hum and CTCSS tones from what listeners hear (section 4.2). Off only to hear the tones themselves, for example to check which one a station sends. It costs the most of the three on the Pi.
Automatic level	on	Evens out quiet and loud stations. Off to hear stations at their real levels, or for an exact node test.
Squelch-tail gate	Light	**Light** mutes the hiss after a station unkeys and long silences. **Strong** also mutes shorter and quieter hiss: use it on a node with noisy links, and turn it back if the start of speech is ever clipped. **Off** passes everything, with no 80 ms delay.

Processing load: how much of one CPU core the cleanup took over the last 5 seconds. On a Pi Zero 2 W it should stay at about 10 % or less while audio passes. **Save** keeps everything on the card; **Reset to 0 dB** puts the gains and buffers back (Save to keep it).

6.6 Microphone card BROWSER

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Leveler	on	A compressor and limiter in the page: quiet speech comes up (by up to 9 dB), loud speech is held back, and nothing passes −1 dBFS. Leave on; turn off only if you level the audio somewhere else.
Automatic gain	off	The browser's own leveling. It fights the leveler and undoes Mic gain, so leave it off when the leveler is on.
Noise suppression	on	The browser removes steady background noise (fans, traffic). Turn off if your voice sounds watery.
Echo cancellation	off	Rarely needed: listen audio is muted while you talk. Turn on only with a speaker right next to the mic.
Voice filter	on	Cuts rumble below about 250 Hz (handling noise, wind, desk thumps).
Radio voice	off	Cuts below 300 Hz and lifts 2.5 kHz for a crisper sound that cuts through on FM. Leave off if your audio already sounds bright, or if the radio adds pre-emphasis.
Noise gate	Off	**Light** opens above −45 dBFS and turns the mic down 15 dB between words; **Strong** opens above −35 dBFS and turns it down 30 dB. Use Light in a noisy room. If the starts of words are clipped, go back a step (the Audio check measures it).
Mic gain	0 dB	−12 to +24 dB. The main control for your level: set it with the mic test (section 7.2).

Test mic and **Test through the node** are on this card; chapter 8 shows their meters and reports.

6.7 Playback card BROWSER

SETTING	DEFAULT	WHAT IT DOES, AND WHEN TO CHANGE IT
Show the spectrum	on	The waterfall on the Listen tab. Off saves battery on an old phone.
Start listening when the page opens	on	Plays as soon as the page loads or you log in. Most browsers need one tap first; the page asks for it.
Keep the screen on	off	Shown where the browser supports it: keeps the screen awake while you listen or talk.
16-bit audio	on	Cleaner on quiet audio, 128 kbit/s instead of 64. Turn off on a slow or metered connection.
Smooth playback	on	Filtered upsampling, no sizzle. Turn off only on a very slow device.

6.8 Log card

What Node Listen has done: listeners, logins, transmissions (with length, gaps and why they ended), connects and every `rpt.conf` edit. Fetched only while the card is open.

Getting levels right

A short procedure for each direction, in the order that avoids fighting yourself. Do it once when you set the node up and again when you change phones, mics or radios.

7.1 Who hears which gain

GAIN	WHERE	WHO IT CHANGES
Radio RX / TX levels, DVSwitch gains	SysMon, Tune tab; DVSwitch	Everyone: the air, every link, Node Listen.
Automatic level, RX gain, Cut below 300 Hz, gate	Node Listen, Audio levels NODE	Node Listen's listeners only.
TX gain	Node Listen, Audio levels NODE	Transmissions from every Node Listen user.
Mic gain, leveler, filters, gate	Node Listen, Microphone BROWSER	Transmissions from this device only.
Volume	Listen tab BROWSER	What you hear on this device only.

TIP Fix things where they start. If the node's own audio is wrong on the air and on links, fix it in SysMon (radio) first; Node Listen's settings only paper over it for Node Listen's users.

7.2 Transmit, step by step

1. **Start clean.** Microphone card: Leveler on, Automatic gain off, Noise suppression on, Voice filter on, Noise gate Off, Mic gain 0 dB.
2. **Test mic.** Hold the phone where you will hold it on the air and talk normally for 4 seconds. Targets: speech **−24 to −12 dBFS**, peaks **−8 to −1 dBFS**, limiter working under 20 % of the time. Change Mic gain by the amount the report suggests and test again until it says *Good*.
3. **Noise.** If the report says *Noisy*, move to a quieter spot or closer to the mic first; then try Noise gate Light.
4. **Test through the node.** For an exact comparison, turn Automatic level off for a minute (Audio levels). *Came back* should be within about 6 dB of *Sent*. If every phone comes back quieter or louder by the same amount, use TX gain; if the radio side is off, use SysMon's Tune tab. Turn Automatic level back on.
5. **On the air.** Run an Audio check during a real contact or net (section 7.4) and apply what it suggests.

7.3 Receive, step by step

1. **The node first.** With the radio's own levels right (SysMon, Tune), the RX level on the Audio levels card should peak around −10 to −3 dBFS on a strong local station.
2. **Look at the spectrum.** A steady line or band below 300 Hz is a CTCSS tone or hum; with Cut below 300 Hz on it should be faint or gone. Thin bright lines right after transmissions are squelch tails: the gate should remove them.

3. **Leave Automatic level on.** If speech is still quiet (the Audio check says so), raise RX gain by the amount it suggests.
4. **Dropouts.** If audio breaks up on your connection, raise the Listen buffer a step.

7.4 The Audio check

The quick tests show four seconds; the Audio check shows a few minutes of real use, with several stations and several of your own transmissions.

1. Setup tab, **Audio check** card: choose a length and press **Start check**.
2. Go to the Listen tab and use the node as usual: listen, talk, listen. The status line shows `CHECK 2:13`; tap it to come back.
3. When it ends (or you press Stop), open the card. Each row is green, amber or red against its target, and **What to change** lists suggestions with amounts.
4. Press **Apply** on a suggestion to make the change (**Undo** puts it back), run another check, and press **Compare with last check** to see the difference.

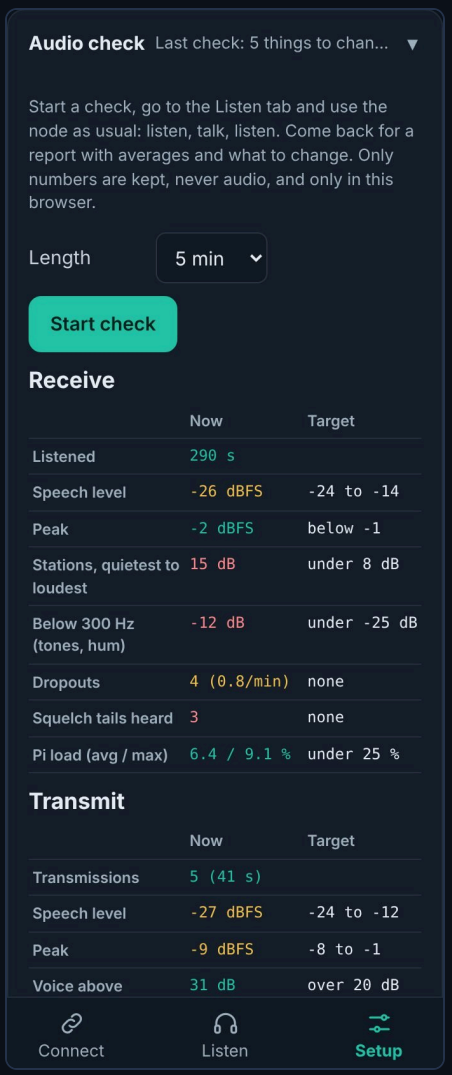


Figure 7.1 · Screenshot (example data). An Audio check report: stations 15 dB apart, hum below 300 Hz, dropouts and squelch tails, each with a suggestion below.

ROW	TARGET	TYPICAL FIX
RX speech level	−24 to −14 dBFS	Automatic level; RX gain
RX peak	below −1 dBFS	Lower RX gain, or Automatic level
Stations, quietest to loudest	under 8 dB	Automatic level
Below 300 Hz	under −25 dB	Cut below 300 Hz
Dropouts	none	Larger Listen buffer; better connection
Squelch tails heard	none	Gate Light, then Strong
Pi load	under 25 %	Cut below 300 Hz off
TX speech level · peak	−24 to −12 · −8 to −1 dBFS	Mic gain
Voice above background	over 20 dB	Noise suppression; Noise gate Light
Limiter working	under 20 %	Lower Mic gain
Gate closed while talking	under 10 %	Noise gate a step lower
Transmissions with gaps	none	Larger TX buffer; Wi-Fi
Peak at the Pi	below −1 dBFS	Lower TX gain

NOTE The check keeps only numbers, never audio, and only in this browser. It carries on if you switch tabs or reload; time with the page closed or the phone asleep is shown as paused.

7.5 Common mistakes

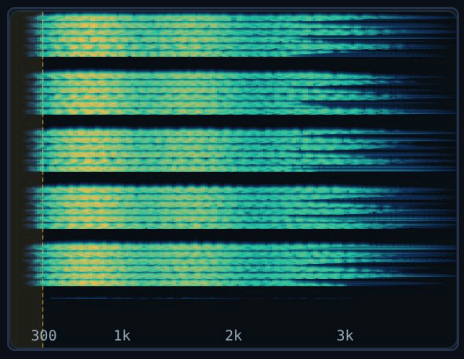
- **Two automatic gains.** The browser's Automatic gain and the leveler fight; leave the browser's off.
- **Mic gain into the limiter.** More mic gain past the point where the limiter works most of the time adds distortion, not loudness. If the report says the limiter works more than 30 % of the time, turn it down.
- **Fixing the radio in Node Listen.** RX gain and Automatic level only change Node Listen's listeners; a bad radio level still goes to every link.
- **Radio voice on a bright radio.** If the radio adds pre-emphasis, Radio voice makes it shrill.
- **Judging levels through the speaker.** Phone volume and speakers vary; trust the meters.

Reading the spectrum and the meters

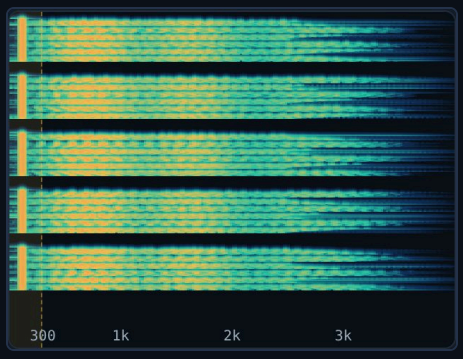
The waterfall shows what the node passes, the meters show levels, and the test reports say what to change.

8.1 The waterfall

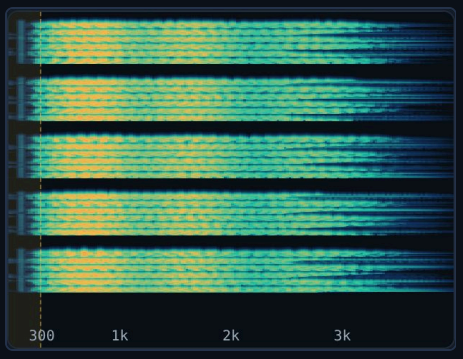
Frequency runs across, 0 to 4 kHz (marks at 300 Hz, 1, 2 and 3 kHz); time runs down, newest at the top, about 25 lines a second. Colour is loudness: dark is quiet, teal is moderate, amber and red are loud. The strip below 300 Hz is shaded. Tap it to switch to bars, which show the same moment as a bar chart. The examples below were made with a synthetic voice on a test node, with each problem added in turn.



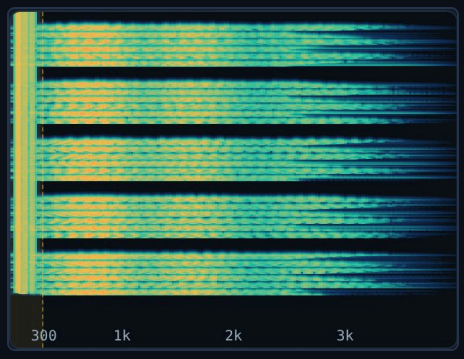
Good speech. Bands of harmonics between 300 Hz and 3.5 kHz, nothing below 300 Hz, clean gaps.



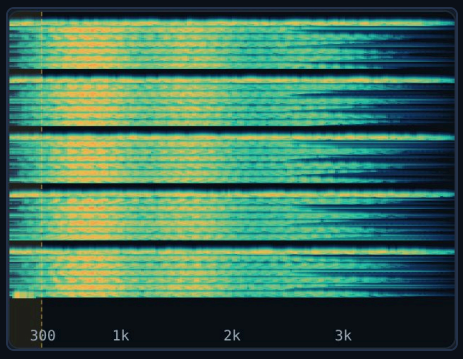
CTCSS tone (100 Hz), cleanup off: a steady bright line in the shaded strip while the station talks.



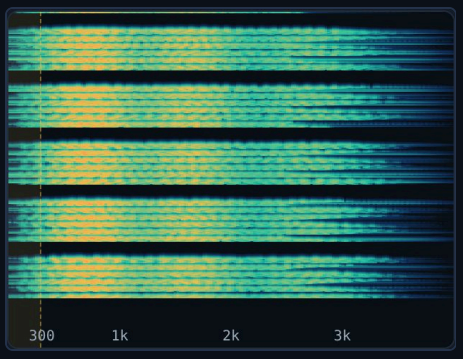
The same with Cut below 300 Hz: the line is gone.



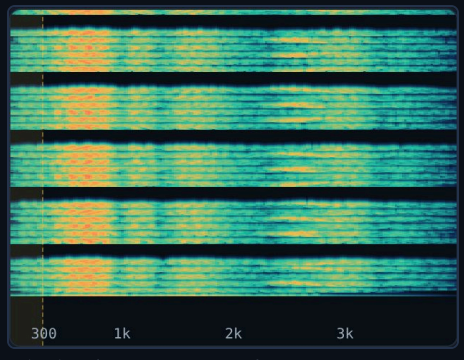
Hum (60, 120, 180 Hz): a band in the shaded strip that carries on between transmissions.



Squelch tails, gate off: a thin bright line across the whole width right after each transmission.



The same with the gate on Light: the tails are gone, the speech is kept.



Clipping (overdriven audio): energy smeared across the band and extra lines that aren't in the voice.

8.2 The live meter and the mic test report

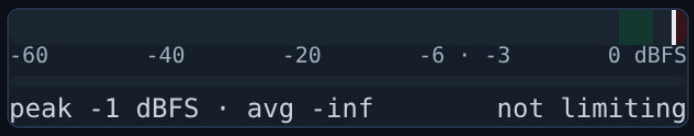


Figure 8.1 · Screenshot. The live meter during a test.

The bar runs from −60 to 0 dBFS. The teal fill is the average, the amber mark the peak, and the thin white mark the highest peak of the last second. The green zone (−6 to −3 dBFS) is where peaks belong; above −1 dBFS is red. The thin bar underneath shows how far the leveler's limiter is turning you down; the text says *limiting 4.2 dB* or *gate closed*.

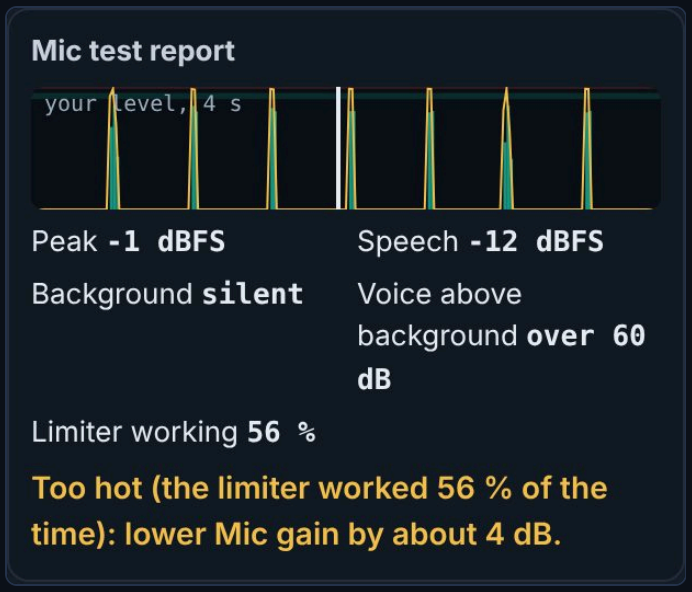


Figure 8.2 · Screenshot (fake test mic). The mic test report: the level over 4 seconds (with a playhead during playback), the numbers, and what to change.

NUMBER	MEANING
Peak	The loudest instant. Should be −8 to −1 dBFS.
Speech	The average while you talk. Should be −24 to −12 dBFS.
Background	The quietest tenth of the recording: the room. *silent* means digital silence.
Voice above background	Speech minus background. Over 20 dB is clean; under 15 dB sounds noisy.
Limiter working	How much of your speech the limiter was turning down. Over 30 %: too much mic gain.
Gate closed while talking	Shown with the noise gate on. Over 10 %: the gate is eating words.

8.3 The node test report

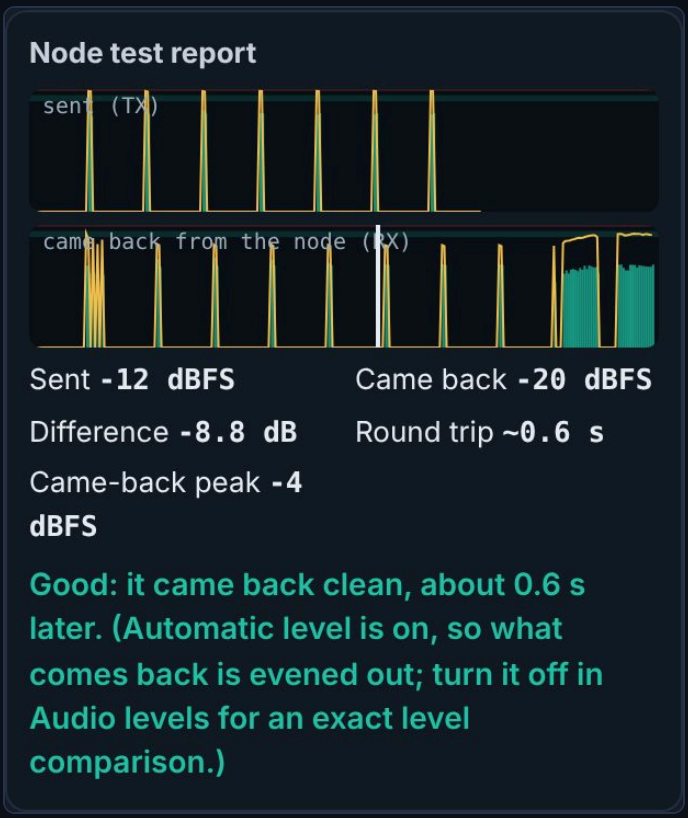


Figure 8.3 · Screenshot (test node). What was sent and what came back, lined up in time.

The test transmits for 4 seconds through the node and records what the node sends back out. The two strips are lined up so the same words sit above each other; **Round trip** is how long that took (about half a second at home). **Difference** compares the speech levels: within about 6 dB is fine.

With Automatic level on, what comes back has been evened out, so the report says so and doesn't judge the difference; turn Automatic level off for a minute to compare exactly.

A *came-back peak* near 0 dBFS means something on the node's side is clipping: TX gain, or the radio and DVSwitch levels in SysMon.

8.4 The level bar while you talk

While you hold push-to-talk, the level bar under the spectrum shows your mic level in red, labelled **TX**; when you let go it goes back to the node's level. Peaks should reach the green zone of the live meter, about −6 to −3 dBFS, on normal speech.

PART III

Everything else

Links and favorites, fixing problems, reaching Node Listen from outside, and the reference tables.

9. Connect and favorites
10. Troubleshooting by symptom
11. From the internet, and security
12. Reference

Connect and favorites

9.1 Favorites

Setup tab, **Favorites** card: up to 10. **Add row**, pick the network (ASL, EchoLink or M17), type the number and a name, and **Save favorites**. **From the Dashboard…** copies one of the Dashboard's favorites.

NETWORK	TYPE
ASL	The node number, up to 7 digits.
EchoLink	The station's own number, up to 6 digits.
M17 TESTING	The reflector and module: `M17-USA C`, or `M17-USA @ALL` (just `M17-USA` means every module).

9.2 Connecting

Connect tab: pick a favorite and press **Connect**. Links are always made from Node Listen's main node. For ASL and EchoLink, Node Listen watches the node's links for up to 15 seconds and says how it went:

RESULT	MEANING
Node 41234 is connected to ASL 2000.	The link is up.
ASL 2000 has not answered after 15 s; node 41234 is still trying.	The far node may be offline or busy.
Node 41234 did not connect to ASL 2000: no link after 15 s.	Check the number; the far node may be refusing connections.

When the main node is also the Dashboard's node and the Dashboard runs, Node Listen asks the Dashboard to connect, so the Dashboard switches to that mode and shows it. M17 favorites always go this way: the Dashboard tunes the M17 bridge and links node 1917 (Dashboard 9.3.92-pi02w or 9.3.88 full and later).

9.3 Links and Disconnect

The Links card at the top of the Connect tab lists the node's links, one per line, with the favorite's name. Node 1921 is marked (TX). **Disconnect** drops linked favorites only; other links stay.

Troubleshooting by symptom

Each from the quickest check to the bigger fix. The Log card and SysMon's Node Listen card (its ASL-DVS tab) show what Node Listen and the node are doing.

No audio at all

1. Status line: *no feed yet* means Asterisk isn't sending. Receive card: is the line on your node? Send its audio here, then Restart Asterisk.
2. Feed green but silent: is the RX light on? Is Listen showing *Listening.*? Check the volume and the phone's own volume.
3. *Tap anywhere to start listening*: the browser is holding sound until you tap.
4. *All 3 listener slots are in use*: raise Max listeners or wait.

Audio breaks up or stutters

1. Raise the Listen buffer a step (0.5 s on mobile data).
2. Turn 16-bit audio off to halve the data.
3. Move closer to the Wi-Fi access point. The Audio check counts dropouts.

Some stations are much quieter than others

Turn Automatic level on (Audio levels). If one station is quiet everywhere, it is that station.

Low hum or a tone under every transmission

Turn on Cut below 300 Hz. If the waterfall shows hum between transmissions too, it is on the node: check the radio interface's grounding.

A burst of hiss after every transmission

Set the Squelch-tail gate to Light, then Strong. The real fix is the radio's squelch or COS on the node sending it.

My transmissions are too quiet, too loud or distorted

Run the mic test and follow it (section 7.2). Distorted with a lot of limiting: lower Mic gain. Fine in the test but bad on the air: the node test, then TX gain or SysMon's radio levels.

Gaps or choppy audio in my transmissions

Raise the TX buffer to 200 ms; use Wi-Fi rather than mobile data. The Log shows gaps per transmission.

Push-to-talk is refused

MESSAGE	FIX
The node is busy	Wait for quiet, or turn off the busy lockout if a constant noise keeps the node busy.
Someone else is transmitting	Wait; the button shows Busy with who and how long.
Node 1921 is not set up / not linked yet	Transmit card: Add node 1921, Restart Asterisk, wait 20 s.
Push-to-talk needs the https:// page	Turn HTTPS on and open the `https://` address.

Microphone: Permission denied	Allow the microphone for the page in the browser's site settings.

Transmission stops by itself

Time-out: raise TX time-out. *Audio stopped reaching the node*: the phone's connection dropped; use Wi-Fi or raise the TX buffer.

Phone-specific

- **iPhone:** sound needs one tap after the page opens; the lock screen keeps playing while the page is the active audio. Push-to-talk stops when the screen locks.
- **Android:** battery savers may stop audio in the background; allow the browser to run in the background.
- **Bluetooth headsets** switch to a low-quality call mode when the microphone opens; expect duller listen audio while you test or talk.

From the internet, and security

Node Listen is meant for your home network. Internet access is off unless you turn it on.

- `allow_wan` in `/etc/asl_dvs/nodelisten.conf` (no checkbox; then `sudo systemctl restart nodelisten`). Off, clients from the internet get *not open to the internet* for every page. Home addresses, link-local and Tailscale (100.64.0.0/10) always count as home.
- A Cloudflare Tunnel or Tailscale Funnel counts as the internet, whatever address it arrives from.
- From the internet the password is only accepted on the `https://` page.
- On top of the per-address lockout, 20 failed internet logins in 10 minutes make every internet login wait until that window ends. Home logins are never held.

> **CAREFUL** The login is the Pi's root password. Before opening Node Listen to the internet, make sure it is long and used nowhere else, and keep the Dashboard and SysMon on the home network.

Reference

12.1 Files

PATH	WHAT
`/usr/local/lib/asl_dvs/nodelisten/`	The program (linked from `/usr/local/bin/nodelisten.py`)
`/etc/asl_dvs/nodelisten.conf`	Node settings (12.4)
`/etc/asl_dvs/nodelisten_favorites.json`	Favorites
`/etc/asl_dvs/nodelisten_tls/`	The HTTPS certificate and key, made on first start
`/etc/asl_dvs/nodelisten_backups/`	`rpt.conf` before each edit, newest 10
`/etc/asl_dvs/local_api.key`	Key for calling the Dashboard

12.2 Ports and service

PORT	WHAT
`8993/tcp`	The page (HTTPS; a plain-HTTP request is redirected)
`6200/udp`	Node audio in, 127.0.0.1 only
`32018 / 34018 udp`	USRP to and from node 1921, 127.0.0.1 only

Service `nodelisten`, through the shared launcher, with `Restart=always` and `WatchdogSec=60`: it stops pinging the watchdog when the page or the audio pump stops answering, and systemd restarts it. `--uninstall` removes the `outstreamcmd` line and node 1921 and keeps the settings; an install within 30 minutes (an update) puts them back.

12.3 Lines Node Listen writes in rpt.conf

```
; nodelisten: sends this node's audio to the Node Listen page (remove it from that page)
outstreamcmd = /usr/bin/python3,/usr/local/bin/nodelisten.py,--feed,6200

; nodelisten: transmit node for the Node Listen page (remove it from that page)
[1921](<your node's template>)
rxchannel = USRP/127.0.0.1:34018:32018
duplex = 0
hangtime = 0
althangtime = 0
holdofftelem = 1
telemdefault = 0
nounkeyct = 1

[nodes]
1921 = radio@127.0.0.1/1921,NONE
```

12.4 Settings keys (nodelisten.conf)

KEY	DEFAULT	RANGE	PAGE
port	8993	1024–65535	Settings: Web port
udp_port	6200	1024–65535	Settings: UDP port
node	(empty)	node number	Receive: Send its audio here
max_listeners	3	1–10	Settings
public_listen	false	bool	Settings: Listen without logging in
https	true	bool	Settings: HTTPS
tot_sec	120	30–600	Settings: TX time-out
busy_lockout	true	bool	Settings
tune_enabled	false	bool	Audio levels: Adjust audio levels here
tx_gain_db / rx_gain_db	0	−12–12	Audio levels
jitter_ms	250	100–1500	Audio levels: Listen buffer
tx_buffer_ms	120	40–400	Audio levels: TX buffer
rx_hpf / rx_agc	true	bool	Audio levels: RX cleanup
rx_gate	1	0 off, 1 light, 2 strong	Audio levels: Squelch-tail gate
allow_wan	false	bool	none (chapter 11)

12.5 API

REQUEST	WHAT
GET /api/health	Version; no login. Used by the watchdog and SysMon.
GET /api/whoami	Logged in?, public listening, Dashboard and SysMon ports; no login.
GET /api/status	Feed, levels, listeners, links, transmit state, Pi load (login, or public listening)
GET /api/listen[?fmt=s16]	The audio stream (login, or public listening)
GET /api/talk	WebSocket for push-to-talk (login, same origin)
GET /api/node · /api/tx · /api/favorites · /api/dash-favorites · /api/config · /api/log	Setup data (login)
POST /api/login · /api/logout	Session

`POST /api/config` · `/api/node/enable` · `/api/node/disable` · `/api/tx/enable` · `/api/tx/disable` · `/api/favorites` · `/api/connect` · `/api/disconnect` · `/api/asterisk/restart`	Changes (login; header `X-Requested-With: XMLHttpRequest`)

12.6 Version history (highlights)

VERSION	WHAT ARRIVED
0.1.0–0.2.0	Listening in the browser through one `outstreamcmd` line
0.3.0	Phone page with tabs, HTTPS, push-to-talk on node 1921, favorites
0.4.0–0.4.5	Tune card (gains, buffers), filtered resampling, Leveler, noise gate, Radio voice, soft limits
0.4.7	Internet access rules (`allow_wan`)
0.4.10	M17 favorites through the Dashboard
0.4.11	Connect reports whether the link came up
0.4.12–0.4.13	Listening starts on open; Setup cards open and close; always the main node
0.4.14	One-screen Listen tab with the waterfall; Status tab removed
0.4.15	RX cleanup: Cut below 300 Hz, Automatic level, Squelch-tail gate
0.4.16	Live meter and reports for the mic and node tests; TX on the level bar
0.4.17	Audio check; the squelch-tail gate tells hiss from voice more reliably

The full list is in `changelogs/nodelisten.md`.